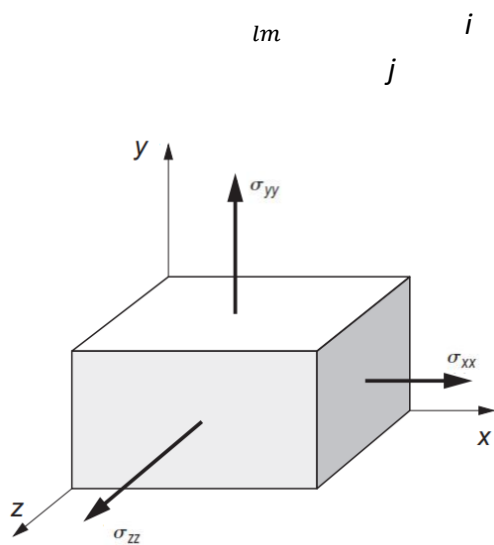
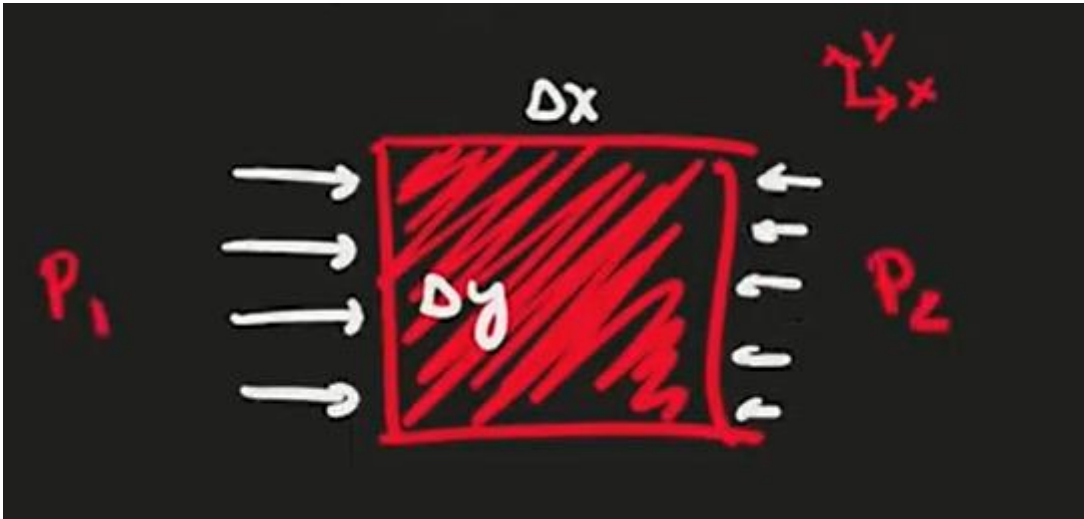




As forças normais ou de pressão





$$s \quad \frac{G}{D} \quad G D$$

$$s \ D \quad s \ D$$

$$D \quad D$$

$$s \quad s$$

$$s \quad s \quad s$$

$$s$$

$$\frac{\tau}{r_0} = \frac{S}{r_0}$$

$$\frac{\tau}{r_0} = \frac{S}{r_0}$$

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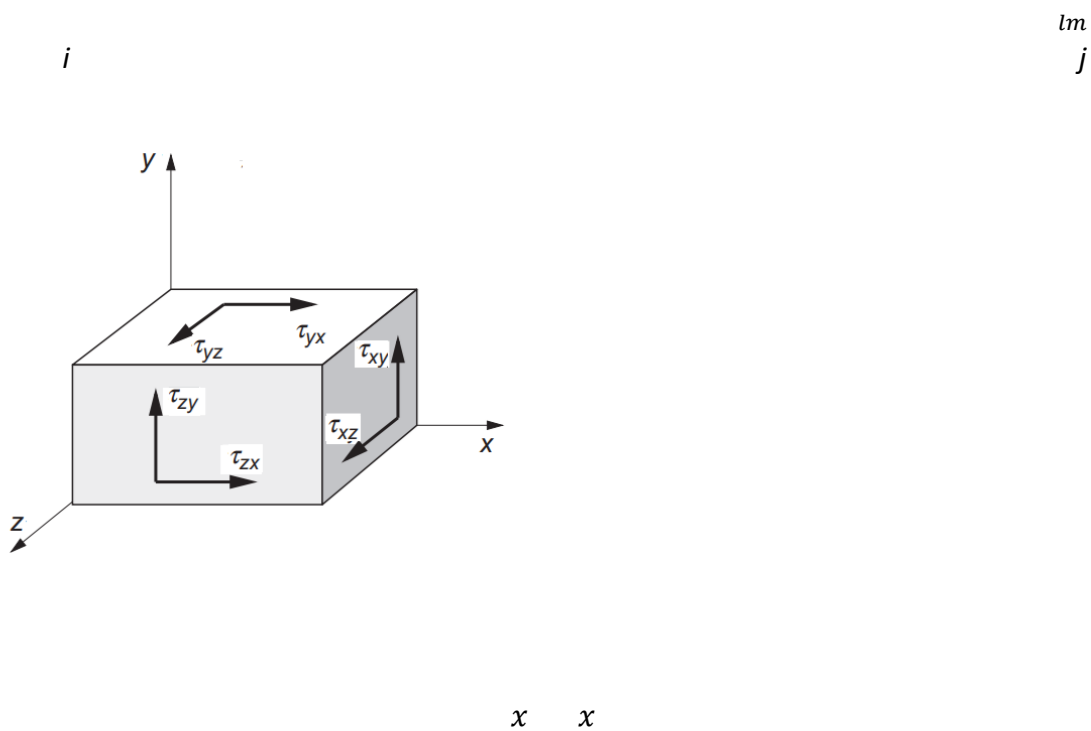
$$\frac{\tau}{r_0} = \frac{S}{r_0}$$

$$\tau \frac{x}{w} = \frac{S}{w}$$

$$\tau \frac{y}{w} = \frac{S}{w}$$

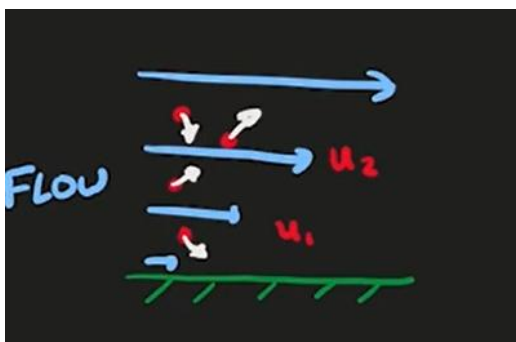
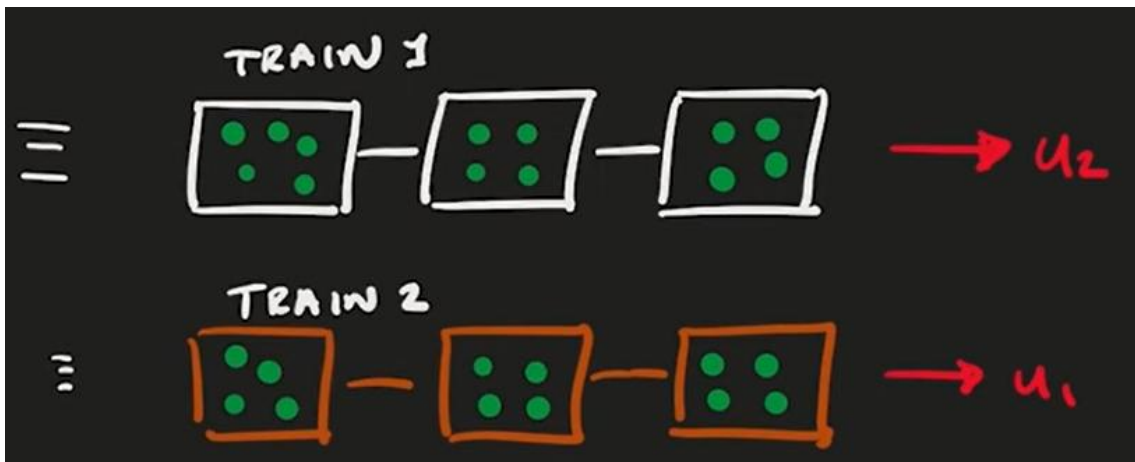
$$\tau \frac{z}{w} = \frac{S}{w}$$

As forças tangenciais, também conhecidas como forças friccionais ou viscosas



x

x



C_D

C_D

$$C_D = \frac{p}{D} \frac{x}{w}$$

$$\frac{x}{w}$$

—

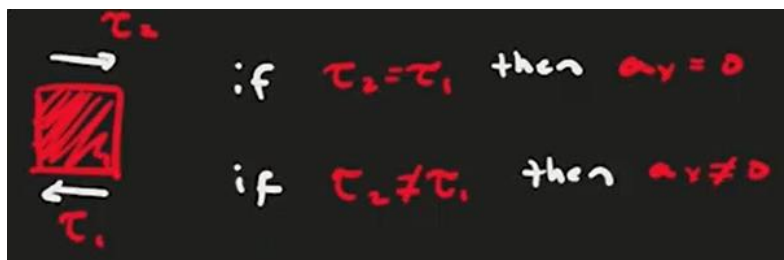
$$\frac{x}{w} = \frac{x}{w} = \frac{x}{w}$$

$$\frac{p}{D} \frac{x}{w} = \frac{p}{D} \frac{x}{w}$$

$$\frac{p}{D} \frac{x}{w}$$

$$\frac{p}{D} \frac{x}{w}$$

$$\frac{x}{w}$$



D D

D D

 x

_____ _____
 x x

 o x

 o x x x

τ
 w s x x x

$$\tau \frac{y}{w} \quad \frac{s}{\quad} \quad \frac{y}{\quad} \quad \frac{y}{\quad} \quad \frac{y}{\quad}$$

$$\tau \frac{z}{w} \quad \frac{s}{\quad} \quad \frac{z}{\quad} \quad \frac{z}{\quad} \quad \frac{z}{\quad}$$

As forças de corpo

$$\mathbb{P} \quad \mathbb{G}$$

$$\text{—————} \quad \mathbb{G}$$

$$\tau \frac{x}{w} \quad \frac{s}{\quad} \quad \frac{x}{\quad} \quad \frac{x}{\quad} \quad \frac{x}{\quad} \quad \tau$$

$$\tau \frac{x}{w} \quad x \frac{x}{\quad} \quad y \frac{x}{\quad} \quad z \frac{x}{\quad} \quad \frac{s}{\quad} \quad \frac{x}{\quad} \quad \frac{x}{\quad} \quad \frac{x}{\quad} \quad \tau \quad C$$

hGr up dGlp lduGdvGluh hvôôôGhu rGhvs hf lydp hq hC

$$\tau \frac{\quad}{w} \quad x \frac{y}{\quad} \quad y \frac{y}{\quad} \quad z \frac{y}{\quad} \quad \frac{s}{\quad} \quad \frac{y}{\quad} \quad \frac{y}{\quad} \quad \frac{y}{\quad} \quad \tau \quad C$$

C

$$\tau \frac{z}{w} \quad x \frac{z}{\quad} \quad y \frac{z}{\quad} \quad z \frac{z}{\quad} \quad \frac{s}{\quad} \quad \frac{z}{\quad} \quad \frac{z}{\quad} \quad \frac{z}{\quad} \quad \tau \quad C$$

C

DvC u vChtxd hvCdflp dCfruhvsrqghp CdGdp rvdC t x rCghCQdylhu V rnhv.Ct xhCqdC
irup dCgh ruldC Cdsuhvhq dgdC r uC

$$\tau \frac{\quad}{w} \quad s \quad \tau \quad C$$

C

C DChlCghCfrqvhuyd rCghCqhuji dC

Srgh vCgylgluCdChqhuji dCghCxp Cvlv hp dChp CgxvCfadvhvCglv lq dvCdChqhuji dC
dup ddhqdgdChCdChqhuji dChp C udqvl rCDsulp hludChv CuhadflrqdgdCgluh dp hq hC
uhadflrqdgdCfrp Cdp dvvdCgrC s ulrCv hp d.ChCqyrCghCvChqhuji dCvCq huqd.Cflq lfdC
hCs r hqfidoCdCvhj xqgdCfadvhCvCuhadflrqdCfrp CdChqhuji dC urfdgdChq uhCv hp dvC
lqfCxlqgrCfCdruCGrCude dkrC r qvlg hudqgrC CdChqhuji dCr dCdup ddhqdgdC r uCxp C
v lv hp dC xhCxp Cq huydCghCp s r.Chfhe hxCp dC xdq lgdghCghCfCdruC Cgh hf x rxC
xp Cude dkrC .Cdf r qvhuyd rCghvvhCv hp dC CudgxdlgdC udy vCgdCh suhv rC

C

DCh suhv rCdf lp dC Cc'sulp hludChlCgdChup rglq p lfd C

q r.CCd dCghCyduld rCgdChqhuji dCs r uCxlgdghCghC hp s rC $\frac{g}{gw}$.Cd vC urfdvCghC
hCqhuji dChih xdgCfrp Cgh hulr uGreGrup dCghCfCdruC $\frac{g}{gw}$.ChCdCghC ude dkr. $\frac{g}{gw}$
CuhvxodqgrCp C

$$\frac{g}{gw} \quad C$$

C xhCvj qlf dCf dhuC xhCd dCghCq udgd.C r uCxlgdghCghCp s r.CghCqhuji dC up lfdC
hCp hf qlfdCxp Cv hp dCp hqr vCfruhvsrqghq hCd dCghCv gdCghyhu Cj xdaudCd dC
ghCf xp xad rCghCqhuji dCqrCq hulr uCghvvhCp hvp rCv hp dC

SdudCuhadflrqduhv hvCfrqfhl rvdCf lp dCfrp Ccqr rCghCyrCp hCghCfrq urdChp suhj d
vhCfrqfhl rCghCghqvlgdghCp vvlfdCghCqhuji dCdup ddhqdgdC hCvCv hp d.ChC

h

μ

$$\frac{g}{g_w} = \frac{h\tau C_{gy}}{w} - h\tau - gDC$$

$$p$$

$$h$$

$$p \quad h$$

$$h$$

$$h \quad s_{uhvv} \, r \quad ylv \, rvdv$$

$$h$$

$$p$$

$$s_{uhvv} \, r$$

$$gD$$

$$g \, s_{uhvv} \, r \quad s \quad - \quad gD$$

$$s_{uhvv} \, r \quad s \quad - \quad gD$$

$$\gamma \rho v dv$$

$$\gamma \rho \int v dv = \rho g D$$

$$\gamma \rho \int v dv = \rho g D$$

$$p + \rho g z = \text{const} = \rho g D$$

$$\frac{1}{2} \rho v^2 + \rho g z = \text{const} = \rho g D$$

Hipóteses comum na mecânica dos fluidos